
State Bottlenecks in Multi-Agent Systems

- 1 Shared context windows explode token costs linearly.
- 2 Cross-agent message passing creates race conditions.

Hierarchical Memory Design

- 1 Short-term memory via scratchpad state buffers.
- 2 Long-term retrieval backed by vector and relational store.



State Sync Implementation

1 Atomic state transitions prevent data race conditions.

2 Sub-15ms sync latency across 8 worker agents.

main.py

```
async def sync_state(agent_id: str, delta: dict):  
    await redis.xadd(agent_stream, delta)
```

Production Takeaways

- 1 Never pass entire raw history between agent nodes.
- 2 Use typed state schemas for deterministic serialization.

